

E.1 / PAPER 1 / QUESTION 28

3 November 2022 / TZ0 / HL Paper 1 / original Q38 / 8822-6501

28. A student quotes three equations related to atomic and nuclear physics:

I. $E = \frac{-13.6}{n^2} \text{ eV}$

II. $N = N_0 e^{-\lambda t}$

III. $mvr = \frac{nh}{2\pi}$

Which equations refer to the Bohr model for hydrogen?

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III